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package np.com.westernit.androidapp;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;
import android.util.Log;
import android.widget.Toast;

import java.util.ArrayList;
import java.util.List;
public class MyDbOpenHelper extends SQLiteOpenHelper {
    public static final String DB_LOG = "DB";
    private static final String DB_NAME = "student.db";
    private static final int DB_VERSION = 1;

    private static final String STUDENT_TABLE_NAME = "tbl_students";
    private static final String ID = "id";
    private static final String FIRST_NAME = "first_name";
    private static final String LAST_NAME = "last_name";
    private static final String STUDENT_TABLE_SQL = "CREATE TABLE " +
STUDENT_TABLE_NAME + "(" +
        ID + " INTEGER PRIMARY KEY AUTOINCREMENT," +
        FIRST_NAME + " TEXT, " +
        LAST_NAME + " TEXT)";

    public MyDbOpenHelper(Context context) {
        super(context, DB_NAME, null, DB_VERSION);
        Toast.makeText(context, "Created", Toast.LENGTH_LONG).show();
        Log.i(DB_LOG, "Database Created Successfully");
    }

    @Override
    public void onCreate(SQLiteDatabase sqLiteDatabase) {
        Log.i(DB_LOG, STUDENT_TABLE_SQL);
        sqLiteDatabase.execSQL(STUDENT_TABLE_SQL);
        Log.i(DB_LOG, "Table Created Successfully");
    }

    @Override
    public void onUpgrade(SQLiteDatabase sqLiteDatabase, int i, int i1) {
        String sql = "DROP TABLE IF EXISTS "+STUDENT_TABLE_NAME;
        sqLiteDatabase.execSQL(sql);
        onCreate(sqLiteDatabase);
    }

    public boolean insert(String firstName, String lastName) {
        ContentValues values = new ContentValues();
        values.put(FIRST_NAME, firstName);
        values.put(LAST_NAME, lastName);

        SQLiteDatabase db = this.getWritableDatabase();
        boolean ins = db.insert(STUDENT_TABLE_NAME, null, values) > 0;
        Log.i(DB_LOG, "Inserted");
        db.close();
        return ins;
    }

    public List<String> readAll() {
        SQLiteDatabase db = getReadableDatabase();
        String sql = "SELECT * FROM " + STUDENT_TABLE_NAME;
        List<String> allStudents = new ArrayList<>();

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        Cursor c = db.rawQuery(sql, null);
        if (c != null) {
            if (c.moveToFirst()) {
                while(!c.isAfterLast()) {
                    String student =c.getString(c.getColumnIndex(FIRST_NAME)) +"
"+c.getString(c.getColumnIndex(LAST_NAME));
                    allStudents.add(student);
                    c.moveToNext();
                }
            }
        }
        db.close();
        return allStudents;
    }

    public boolean updateRecord(long id,String firstName, String lastName)
    {
        SQLiteDatabase db = getWritableDatabase();
        ContentValues args = new ContentValues();
        args.put(ID, id);
        args.put(FIRST_NAME, firstName);
        args.put(LAST_NAME, lastName);
        int i = db.update(STUDENT_TABLE_NAME, args, ID + "=" + id, null);
        return i > 0;
    }

    public boolean deleteRecord(Integer id){
        SQLiteDatabase db = getWritableDatabase();
        // db.execSQL("DELETE FROM " + STUDENT_TABLE_NAME+ " WHERE "+ID+"='"+id+"'");
        return db.delete(STUDENT_TABLE_NAME, ID + "=" + id, null) > 0;

        //
    }
}

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package np.com.westernit.androidapp;

import android.os.Bundle;
import android.support.v7.app.AppCompatActivity;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;

public class DbActivity extends AppCompatActivity {
    private MyDbOpenHelper db;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_db);

        db = new MyDbOpenHelper(this);
    }
}

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        if(db.updateRecord(3,"Rohit","Rimal")){
            Toast.makeText(getApplicationContext(),"Record
Updated",Toast.LENGTH_LONG).show();
        }else {
            Toast.makeText(getApplicationContext(),"Sorry unable to
Updated",Toast.LENGTH_LONG).show();
        }

        Button btnCreate = findViewById(R.id.createRecord);
        final Button btnRead = findViewById(R.id.readRecord);

        btnCreate.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                EditText etFirstName = findViewById(R.id.et_first_name);
                EditText etLastName = findViewById(R.id.et_last_name);

                if(db.insert(etFirstName.getText().toString(),
etLastName.getText().toString())){
                    Toast.makeText(getApplicationContext(),"Record
Saved",Toast.LENGTH_LONG).show();
                }else {
                    Toast.makeText(getApplicationContext(),"Unable to Save
Record",Toast.LENGTH_LONG).show();
                }
            }
        });

        btnRead.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                TextView tvDbResult = findViewById(R.id.db_result);
                StringBuffer buffer = new StringBuffer();
                for(String s: db.readAll()){
                    buffer.append(s+"\n");
                }
                tvDbResult.setText(buffer);
            }
        });
    }
}

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<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".DbActivity"
    android:orientation="vertical">

    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/et_first_name" />

    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"

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        android:id="@+id/et_last_name" />
```

```
<Button
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/createRecord"
    android:text="Create Reccord"/>
```

```
<Button
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:id="@+id/readRecord"
    android:text="Read Record"/>
```

```
<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_below="@+id/readRecord"
    android:id="@+id/db_result"/>
```

```
</LinearLayout>
```